

Kuhn-Tucker Conditions ^{*}

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September 15, 2011

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^{*}Comments are welcome. Please refer to Dixit[1] and MWG[2] for further discussions.

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1 Nonlinear Programming Problem

A standard nonlinear programming problem is

$$\begin{aligned} & \max_{x \in \mathbb{R}^N} f(x_1, \dots, x_N) \\ & \text{s.t.} \quad \begin{cases} g_1(x_1, \dots, x_N) = b_1 \\ \dots \\ g_M(x_1, \dots, x_N) = b_M \\ h_1(x_1, \dots, x_N) \leq c_1 \\ \dots \\ h_K(x_1, \dots, x_N) \leq c_K \end{cases} \end{aligned} \tag{1}$$

2 Kuhn-Tucker Conditions

Theorem (*Kuhn-Tucker Conditions*) Suppose that $x^* = (x_1^*, \dots, x_N^*)$ solves problem (1) where f and $g_1, \dots, g_M$ are C^1 functions. Assume that the constraint qualification¹ is satisfied. Then there are multipliers $\lambda_m \in \mathbb{R}$, with $m = 1, \dots, M$, and $\lambda_k \in \mathbb{R}_+$, with $k = 1, \dots, K$, such that:

1. For every $n = 1, \dots, N$,

$$\frac{\partial f(x^*)}{\partial x_n} = \sum_{m=1}^M \lambda_m \frac{\partial g_m(x^*)}{\partial x_n} + \sum_{k=1}^K \lambda_k \frac{\partial h_k(x^*)}{\partial x_n}$$

2. For every $k = 1, \dots, K$,

$$\lambda_k (h_k(x^*) - c_k) = 0$$

which are also called *Complementary Slackness* conditions.

Let us define the constraint set $\mathbb{C}$ as

$$\mathbb{C} \equiv \{x \in \mathbb{R}^N : g_m(x) = b_m, h_k(x) \leq c_k, \text{ for } m = 1, \dots, M, k = 1, \dots, K.\}$$

then we have the following theorem.

Theorem (*Uniqueness*) Suppose that in problem (1) the constraint set $\mathbb{C}$ is convex and the objective function $f(\cdot)$ is strictly quasiconcave. Then there is a unique global constrained maximizer.

¹The constraints are independent at x^* .

3 Examples

Example 1

Use the Kuhn-Tucker condition to find the optimal solution in

$$\begin{aligned} \min_{x,y \in \mathbb{R}} \quad & e^{-x} + e^{-y} \\ \text{s.t.} \quad & x^2 + y^2 \leq 2 \end{aligned}$$

Example 2

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 - 2x_2 \\ \text{s.t.} \quad & -1 - x_1 + x_2^2 \leq 0 \\ & -x_2 \leq 0 \end{aligned}$$

Example 3

$$\begin{aligned} \max_{x,y} \quad & U = xy \\ \text{s.t.} \quad & 100 \geq x + y \\ & x \leq 40 \end{aligned}$$

Example 4

With n goods, choose $(x^a, x^b) \in \mathbb{R}_{++}^n \times \mathbb{R}_{++}^n$ to solve the problem

$$\begin{aligned} \max_{(x^a, x^b)} \quad & u^a(x^a) \\ \text{s.t.} \quad & x^a + x^b = \bar{x} \\ & u^b(x^b) = \bar{u}^b \end{aligned}$$

References

- [1] Dixit, A. 1990, *Optimization in Economic Theory*, 2nd Edition, Oxford University Press.
- [2] Andreu Mas-Collel, Michael Whinston, and Jerry R. Green, 1995, *Microeconomic Theory*, Oxford University Press.